```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
```

```
df = pd.read_csv("/content/diploma_recognition.csv")
```

```
df = df.drop(columns=["reference", "note", "sources", "confidence_detail"])
```

```
target = "confidence"
```

```
for column in df.columns:
      if df[column].dtype == "object":
              le = LabelEncoder()
              df[column] = le.fit_transform(df[column].astype(str))
```

```
df = df.fillna(df.median(numeric_only=True))
```

```
X = df.drop(columns=[target])
y = df[target]

print("Feature Names:", list(X.columns))
print("Target Classes:", y.unique())
print("Number of Samples:", X.shape[0])
print("Number of Features:", X.shape[1])

print(df.head())

print("\nClass Distribution")
print(y.value_counts())
```

```
Feature Names: ['country_or_system', 'certificate', 'uk', 'us', 'germany', 'index_vs_cam
Target Classes: [0 1]
Number of Samples: 76
Number of Features: 12
   country_or_system  certificate   uk   us  germany  \
0                 34           58  1.0  0.7      0.7
1                 62           60  1.0  0.7      1.0
2                  5           11  1.0  0.7      1.0
3                 72            0  1.0  1.0      0.7
4                 18           55  1.0  0.7      1.0

   index_vs_cambridge_a_level  prestige_uk_top5  oxford_status  \
0                        0.81               1.0              0
1                        1.00               1.0              0
2                        1.00               1.0              0
3                        1.00               1.0              0
```

```
4                        1.00               1.0               0

   cambridge_status  imperial_status  ucl_status  lse_status  confidence
0                 0                0           0           0           0
1                 0                0           0           0           1
2                 0                0           0           0           1
3                 0                0           0           0           0
4                 0                0           0           0           0

Class Distribution
confidence
1    40
0    36
Name: count, dtype: int64
```

```
X_train, X_test, y_train, y_test = train_test_split(
      X,
      y,
      test_size=0.2,
      random_state=0,
      stratify=y
)

print("Shape of X_train:", X_train.shape)
print("Shape of X_test:", X_test.shape)
```

```
Shape of X_train: (60, 12)
Shape of X_test: (16, 12)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)

print("Model Trained Successfully!")
```

```
Model Trained Successfully!
```

```
y_pred = model.predict(X_test)
```

```
results = pd.DataFrame({
      "Actual": y_test.values,
      "Predicted": y_pred
})

print(results.head(10))
```

```
   Actual  Predicted
0       1          1
1       1          0
2       0          1
3       0          1
4       0          1
5       0          1
```

```
6       0          1
7       1          1
8       1          1
9       1          1
```

```
accuracy = accuracy_score(y_test, y_pred)

print(f"\nTest Accuracy: {accuracy:.2%}")
```

```
Test Accuracy: 56.25%
```